

Graphics on Linux

Kapil Agrawal, designer and CEO, Mediamagic Technologies, shares, "Earlier, it was very difficult to utilise all the resources of a GPU inside Linux, as there were neither proper APIs to offload hardware acceleration needed by video encoding/decoding, nor enough tools from GPU companies as they always saw non-Linux as the biggest market."

"But, with many users and companies moving towards Linux, tools are now available to do software development on GPUs in Linux. Also, GPU companies have released APIs that can now directly talk with the GPU for hardware acceleration, like VDPAAU from NVIDIA, VAAPI from Intel and XVBA from ATI."

is a significant improvement in data transfer speed. Most of the products in this category have a significant cache size on the processor. Now a graphics processor can take advantage of this high-speed cache available right next to it. The input/output traffic flow improves quite a bit, as the traffic moves from the motherboard to the die itself.

Bhandari explains, "It's analogous to the situation where you are running a factory. You are doing the computing there, taking the assembled goods, putting them in a truck, sending them to a separate graphics factory, getting the processing done and bringing the goods back. Now you are doing it inside the factory right next to your assembly line."

So the transfer time for the data back from the memory is greatly reduced.

Power management is another important advantage. For high-performance tasks like music ripping or video compression-decompression, the hardware works hard for a short duration of time and maintains its low-power state for the rest of the time.

Fusion GPUs accelerate processing

While Intel tries to put the CPU and GPU on the same socket for a low-level integration, the accelerated processing unit (APU) from AMD provides a higher level of integration. An APU is basically an x86 CPU combined with memory controller and graphics processor.

One major difference between discrete graphics chips and APUs is that in discrete graphics chips the entire silicon and all of its transistors face pixel-crunching tasks, whereas in APUs the technology bandwidth is divided

for the rest of the computer—CPU and memory controller.

The Fusion technology promises performance at par with some of the discrete graphics chips. However, it cannot be compared or serve as an alternative to high-end discrete graphics chips for serious gamers.

One major advantage of Fusion approach in APUs is its minimal access times and maximum bandwidth. The introduction of APUs does not mean a replacement of the CPU in the design.

Vamsi Krishna, manager-product marketing APAC, AMD, explains, "CPUs will still be the driving brain. Without the classic CPUs you cannot have the millions of applications running on your product."

In the performance desktop space, discrete graphics chips will still lead the graphics computing power. He says, "The bulk of general-purpose computing will move towards APUs giving you the advantage of integration, price and real estate."

Intensive computing applications

Today, graphics cards not only provide the best graphics but also support applications like video transcoding, video acceleration, facial recognition and facial tagging where the data is massive and needs parallel processing capabilities.

Krishna says, "GPUs are becoming more and more generic processors than what was intended before. While the primary objective is graphics, people have started extracting the secondary objective from it."

3D home entertainment to personal computing. Graphics processors have

become central to almost every aspect of computing and this is expected to be a long-term trend.

Goyal informs, "Stereoscopic 3D is a huge trend right now. Within less than a year, 3D has gathered incredible global momentum. The whole industry ecosystem has very much embraced the technology. 3D-capable panels from LG, Samsung, Viewsonic, Alienware and Acer, and 3D Vision enabled desktop and notebook PCs from major OEMs are evidence of this. The advent of 3D Blu-ray and YouTube confirms that 3D is much more than a gimmick—it's here to stay."

Another noteworthy trend is seen in professional graphics. There has been an inflection point with the emergence of 'computational visualisation' era. For high-precision, data-sensitive applications, professional graphics solutions with error-correcting code memory and fast double-precision capabilities ensure the accuracy and fidelity of your results. These not only serve as a graphics processor but also drive an entire visual supercomputing platform, incorporating hardware and software that enable advanced capabilities such as stereoscopic 3D, scalable visualisation and 3D high-definition broadcasting.

Multi-display solutions. Buying a large-screen panel may be very expensive, but it's now easy to stretch the video into three or six smaller screens without compromising on the quality of video.

Krishna adds, "The efynity feature enables you to view the video split into smaller screens joined together to form a large screen, so that you can enjoy high-end gaming."

"The way you have USB port hubs, you now get display port hubs. With one hub, you can connect up to four monitors, take a display port connector and expand it to four monitors. This will give an altogether different game play experience. Both the gamers and productivity guys can take advantage of this, especially if they have to work on multiple monitors."

Video stabilisation. Krishna shares